

Sharp Sobolev Extension Constant for Planar Sectors: An Exact Later Answer to the Wedge Asymptotic

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Status and sources

Status: literature_implied_answer (full).

The original problem is Remark 4.6 on PDF page 15 of Lotoreichik's arXiv:1306.1524 [1]. The exact answer is Theorem 1 on PDF page 2 of Zeraoulia's June 2026 preprint [2], deposited at Zenodo under DOI 10.5281/zenodo.20534573. The later paper does not cite [1]; the answer relation follows by matching the definitions and theorem, hence the provenance classification.

Source problem

For a planar wedge Ω_φ of opening $0 < \varphi \leq \pi$, let

$$\mathcal{E}(\Omega_\varphi) = \inf_E \|E\|_{H^1(\Omega_\varphi) \rightarrow H^1(\mathbb{R}^2)}, \quad (Ef)|_{\Omega_\varphi} = f,$$

where

$$\|f\|_{H^1(D)}^2 = \int_D (|f|^2 + |\nabla f|^2) dx.$$

The source proves, with $\theta \downarrow 0$,

$$\sqrt{2} + \frac{\sqrt{2}}{32}\theta^2 + o(\theta^2) \leq \mathcal{E}(\Omega_{\pi-\theta}) \leq \sqrt{2} + \frac{\sqrt{2}}{4}\theta + o(\theta),$$

and asks for the exact asymptotics because the displayed bounds even have different orders.

Exact answering theorem

Put $\beta = 2\pi - \varphi$. Theorem 1 of [2] computes the exterior and full-plane sharp constants as

$$C_{\text{out}}(\varphi)^2 = \max \left\{ \frac{\varphi}{\beta}, \frac{\beta}{\varphi} \right\}, \quad C_{\text{full}}(\varphi)^2 = 1 + C_{\text{out}}(\varphi)^2 = \frac{2\pi}{\min\{\varphi, \beta\}}.$$

Its C_{full} is exactly the source's $\mathcal{E}$: both include the fixed H^1 contribution on the original sector as well as the added exterior part. Therefore, for $0 < \varphi \leq \pi$,

$$\boxed{\mathcal{E}(\Omega_\varphi) = \sqrt{\frac{2\pi}{\varphi}}}.$$

Substituting $\varphi = \pi - \theta$ gives the complete answer

$$\mathcal{E}(\Omega_{\pi-\theta}) = \sqrt{2} \left(1 - \frac{\theta}{\pi}\right)^{-1/2} = \sqrt{2} + \frac{\sqrt{2}}{2\pi}\theta + \frac{3\sqrt{2}}{8\pi^2}\theta^2 + O(\theta^3).$$

In particular, the first correction is linear, with sharp coefficient $\sqrt{2}/(2\pi)$.

Why the sharpness proof matches the source norm

The upper operator on the complementary sector is the angular reflection–dilation

$$(T_0 u)(r, \eta) = u\left(r, \frac{\varphi(2\pi - \eta)}{\beta}\right), \quad \varphi < \eta < 2\pi.$$

A polar-coordinate change of variables gives

$$\|T_0 u\|_{H^1(\Omega_\varphi^c)}^2 = \frac{\beta}{\varphi} \int_{\Omega_\varphi} (|u|^2 + |u_r|^2) + \frac{\varphi}{\beta} \int_{\Omega_\varphi} \frac{|u_\vartheta|^2}{r^2},$$

so the exterior upper constant is the stated maximum.

For sharpness, $t = \log r$ transforms the Dirichlet energy conformally into the standard energy on an infinite strip. Slowly varying equal boundary traces force the ratio β/φ ; slowly varying opposite traces force φ/β . The Fourier strip calculation is on PDF pages 3–5 of [2]. Finally, spatial dilation leaves the two-dimensional Dirichlet energy unchanged while multiplying the L^2 term by ρ^2 ; letting $\rho \downarrow 0$ proves that the inhomogeneous H^1 constant cannot be smaller. Thus no homogeneous/inhomogeneous or exterior/full-plane convention gap remains.

References

- [1] V. Lotoreichik, *Lower bounds on the norms of extension operators for Lipschitz domains*, arXiv:1306.1524 (2013).
- [2] R. Zeraoulia, *Sharp $W^{1,2}$ Extension Constants for Planar Sectors*, Zenodo, 4 June 2026, DOI: 10.5281/zenodo.20534573.